ADVANCED INDUSTRIAL ORGANIZATION

LECTURE 1

Firms, Consumers and the Market

Costs, incentives, demand and market benchmarks

Study notes based on the lectures of Hui Shen

Xucheng CHEN

The Chinese University of Hong Kong, Shenzhen

October 2026

Reading guide

This is a self-contained set of notes for Lecture 1 of Advanced Industrial Organization, based on Hui Shen's annotated slides dated September 4, 2026. The lecture sequence is preserved, while progressive slide builds are developed into connected explanations. Core material, cases, numerical tables, starred topics and relevant appendix proofs are included. Mathematical extensions fill in intermediate steps and clarify the conditions behind the results.

Read each model in the same order: **environment** $\longrightarrow$ **choice problem** $\longrightarrow$ **first-order condition** $\longrightarrow$ **equilibrium** $\longrightarrow$ **economic interpretation**. Before comparing two results, check which action or market condition is held fixed.

Definitions and core results

Blue boxes collect the definitions and conclusions to retain. The text around them supplies the assumptions and the derivation.

Economic intuition

Green boxes identify the economic mechanism and explain why a change in incentives affects the outcome.

Examples and lecture cases

Purple boxes connect a model to a calculation or an application. Case figures and events from earlier periods are treated as historical illustrations.

Conditions and distinctions

Amber boxes highlight interior-solution requirements, boundary cases and distinctions needed to interpret the result correctly.

The text states corrected results directly. Source references use physical PDF pages, because the annotated file contains repeated slide numbers and progressive builds. The source concordance at the end makes it possible to return to the corresponding classroom material.

Notation

Symbol	Meaning
q_i, $Q = \sum_i q_i$	Firm output and total industry output.
$Q_i(\boldsymbol{p})$, $P(Q)$	Direct demand and inverse market demand.
$C_i(q_i)$, $MC_i = C_i'(q_i)$	Total economic cost and marginal cost.
π_i	Profit; specify whether committed fixed costs are included.
$s_i = q_i/Q$, $H = \sum_i s_i^2$	Market share and normalized Herfindahl index.
$\eta = -Q'(p)p/Q(p) > 0$	Absolute price elasticity of demand.
τ, r_i, M	Mismatch/transport cost, valuation and consumer mass.
a, b, d	Demand intercept, own-quantity coefficient and cross-quantity coefficient.
λ, γ	Conduct parameter and conjectural variation.
$R_i(x_j)$	Best response of firm i to a rival's action.
$[z]_+ = \max\{z, 0\}$	Positive part, used for nonnegative demand or output.
CS, TS	Consumer surplus and total surplus.

Three distinctions to keep visible. A product's substitutability is a feature of demand; strategic substitutability is a feature of incentives. A fixed cost is independent of current output; a sunk cost cannot be recovered. A first-order condition is an optimization condition; equilibrium also requires that all agents' choices fit together and that their constraints are respected.

Contents

Reading guide i

Notation ii

1 The Firm: Costs and Objectives 1

1.1 Economic costs are opportunity costs . . . 1
1.2 Scale, scope and the shape of costs . . . 1
1.3 Fixed costs, sunk costs and decision timing . . . 2
1.4 Profit maximization as a benchmark . . . 3

2 Inside the Firm: Incentives and Boundaries 4

2.1 Why the black-box firm may fail . . . 4
2.2 A moral-hazard model . . . 4
2.3 Observable effort: why full insurance is optimal . . . 5
2.4 Hidden effort and a risk-neutral manager . . . 5
2.5 The opposite polar case: extreme risk aversion . . . 6
2.6 The boundaries of the firm . . . 6

3 Consumers and Demand 7

3.1 Behavioral assumptions . . . 7
3.2 From utility to a demand function . . . 7
3.3 A linear-quadratic model with two goods . . . 8
3.4 Individual and aggregate demand . . . 9

4 Welfare in a Partial-Equilibrium Model 10

4.1 What is held fixed? . . . 10
4.2 Consumer surplus for one good . . . 10
4.3 Consumer surplus with several goods . . . 11
4.4 Producer surplus, total surplus and distribution . . . 11

5 Competition and Monopoly 13

5.1 The perfectly competitive benchmark . . . 13
5.2 A monopolist's choice . . . 14
5.3 Multiproduct monopoly: the general condition . . . 15

6 A Dominant Firm and a Competitive Fringe **17**

6.1 The economic idea and the lecture case 17

6.2 Hotelling demand with two endpoints 17

6.3 Constant marginal costs 18

6.4 Increasing marginal costs: the fringe reacts 18

7 Market Definition and Market Performance **20**

7.1 From isolated decisions to strategic interaction 20

7.2 Demand substitution and the hypothetical monopolist 20

7.3 Supply substitution and entry 21

7.4 The Lerner index 21

7.5 Concentration ratios and the Herfindahl index 21

7.6 Lecture case: concentration across manufacturing industries 23

A A Compact Formula Reference **24**

B Source Concordance **25**

Sources and Further Reading **27**

CHAPTER 1

The Firm: Costs and Objectives

Lecture 1 · source PDF pp. 1–27

Industrial organization studies how firms interact in markets and how their choices affect consumers and welfare. The first step is to specify the two sides of a firm's profit: the cost of supplying output and the revenue generated by demand. Technology and input prices determine costs; consumer preferences and market interaction determine revenues. Initially, the firm is treated as a single decision-maker. Chapter 2 opens this black box.

1.1 Economic costs are opportunity costs

The relevant cost of an action is the value of the best alternative forgone. An accounting payment, an historical expenditure and an economic cost need not be the same. A factory that the firm already owns may require no new cash payment, yet using it can sacrifice rental income or the profit from another product. Conversely, an irrecoverable past expenditure does not become relevant merely because it appears in the accounts.

Suppose a capacity unit can produce either a new product or an existing product that would earn a contribution of \$20 after its own variable inputs. If the new product uses \$8 of additional inputs, its economic incremental cost is \$28: \$8 paid now plus \$20 forgone. This is why observed cost data require interpretation before they can be used to infer market power.

A cost function is itself the outcome of optimization. With input vector z, input prices w and technology f, write

$$C(q;w) = \min_{z \geq 0} \{w \cdot z : f(z) \geq q\}. \tag{1.1}$$

It gives the *minimum economic cost* of producing q at given technology and input prices. In most models here, C_i is independent of rivals' current actions. This is an assumption: common input markets, congestion or learning spillovers could make such independence inappropriate.

1.2 Scale, scope and the shape of costs

1.2.1 Economies of scale

For a single product, average cost is $\mathrm{AC}(q) = C(q)/q$ and marginal cost is $\mathrm{MC}(q) = C'(q)$. Differentiating average cost gives

$$\mathrm{AC}'(q) = \frac{qC'(q) - C(q)}{q^2} = \frac{\mathrm{MC}(q) - \mathrm{AC}(q)}{q}. \tag{1.2}$$

Thus average cost falls when marginal cost lies below average cost, and rises when marginal cost lies above it.

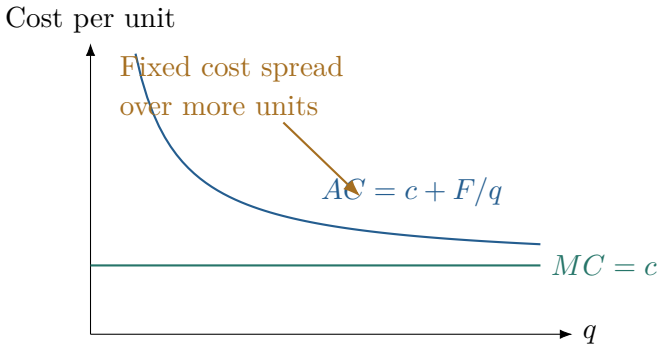
Scale economies

A firm has economies of scale over a range if average cost falls with output. It has diseconomies of scale if average cost rises. With $C(q) = F + cq$ for $q > 0$,

$$\mathrm{MC}(q) = c, \qquad \mathrm{AC}(q) = c + \frac{F}{q}, \qquad \mathrm{AC}'(q) = -\frac{F}{q^2} < 0.$$

Constant marginal cost and positive fixed cost can therefore coexist with economies of scale.

Scale economies favor concentration because spreading fixed costs over a larger quantity can make a large producer cheaper than several small ones. This is the connection to the course's earlier Alcoa example; that earlier case is referenced, rather than reproduced, in the supplied slides. Technological congestion and the cost of managerial oversight can eventually create diseconomies. Marginal cost itself can be constant, increasing, decreasing or U-shaped; scale economies concern *average* cost.



1.2.2 Economies of scope

For two products, a standard cost comparison is

$$C(q_1, q_2) < C(q_1, 0) + C(0, q_2). \tag{1.3}$$

Joint provision then saves resources relative to separate provision. Shared distribution, advertising, a production facility or a management system can generate such economies. The reverse inequality describes diseconomies of scope.

A related *local* concept is cost complementarity:

$$C_{12}(q_1, q_2) = \frac{\partial^2 C}{\partial q_1 \partial q_2} < 0. \tag{1.4}$$

Producing more of product 2 then lowers the marginal cost of product 1. This is the local link used in the multiproduct pricing analysis. Global joint-cost savings and a negative cross-partial are related concepts, but are not identical definitions; a shared fixed cost can generate scope economies without changing marginal costs.

1.3 Fixed costs, sunk costs and decision timing

A *fixed cost* does not vary with current output within a specified operating regime and time horizon. A *sunk cost* cannot be recovered when a past decision is reversed. The distinction

concerns two different questions: whether cost changes with output, and whether cost can still be avoided or recovered.

Examples of sunk expenditures include an irrecoverable shop refurbishment, a nontransferable business licence and some entry advertising. The required advertising may be endogenous to incumbents' reputation: entering a market with established brands can require a larger initial campaign. Sunk costs affect entry, exit and industry structure even when they disappear from a short-run pricing first-order condition.

Marginal decisions versus participation decisions

Once an unavoidable fixed cost is committed, it does not enter the first-order condition for current output or price. An *avoidable* fixed operating cost still matters for whether to operate at all. A sunk entry payment is irrelevant to the current choice but was relevant before entry. Always specify the decision date.

For example, with avoidable operating cost F and variable cost $VC(q)$, operating is worthwhile only if

$$\max_{q>0}\{pq - VC(q)\} \geq F.$$

If F is already unavoidable, compare variable revenue with variable cost instead. This distinction also explains why charging an arbitrary percentage over average total cost need not maximize profit: the optimal price follows from marginal incentives and demand. Cost-plus pricing can coincide with an optimum in particular cases, but it is not a general pricing rule.

1.4 Profit maximization as a benchmark

For a monopolist, profits are

$$\pi(q) = qP(q) - C(q). \tag{1.5}$$

In an oligopoly, the demand a firm faces also depends on rivals' actions. Advertising, research and development, quality and organizational choices can shift demand or costs. The simple output problem is a starting point, not a claim that these choices do not exist.

Profit maximization is a useful benchmark for firms with market power and for competition analysis. Owners may additionally care about a media outlet's political influence, revenue, market share, local employment or customers' well-being. Even if owners maximize profits, hired managers may not share that objective. This motivates the next chapter: when does an incentive contract make the firm behave as if it maximized a single profit function?

CHAPTER 2

Inside the Firm: Incentives and Boundaries

Lecture 1 · source PDF pp. 28–57

2.1 Why the black-box firm may fail

An organization contains agents with different objectives and information. Top managers may pursue empire building, including mergers that increase their status without increasing shareholder value. Middle managers may optimize their own divisions rather than the whole firm. Delegation nevertheless has benefits because information is dispersed and senior managers have limited attention. Strategic communication can further affect which information reaches decision-makers.

We first isolate the owner–manager problem from product-market competition. The purpose is to understand whether performance pay can align their incentives.

2.2 A moral-hazard model

Let gross profit be $\Pi(e, \varepsilon)$, where $e \in [\underline{e}, \bar{e}]$ is the manager's effort and ε is a random shock. Higher effort raises gross profit for a given shock. The risk-neutral owner observes realized profit but cannot directly contract on hidden effort, and offers compensation $w(\Pi)$. The manager's utility satisfies

$$u_w > 0, \qquad u_e < 0, \qquad u_{ww} \leq 0.$$

The manager likes pay, dislikes effort and is weakly averse to income risk. Her reservation expected utility is u_0. The owner chooses a contract and an induced effort level to solve

$$\max_{w(\cdot), e^*} \quad \mathbb{E}[\Pi(e^*, \varepsilon) - w(\Pi(e^*, \varepsilon))] \tag{2.1}$$

$$\text{subject to} \quad \mathbb{E}[u(w(\Pi(e^*, \varepsilon)), e^*)] \geq u_0, \tag{2.2}$$

$$e^* \in \arg\max_{e} \mathbb{E}[u(w(\Pi(e, \varepsilon)), e)]. \tag{2.3}$$

Equation (2.2) is *individual rationality* or participation: the manager must prefer accepting to her outside option. Equation (2.3) is *incentive compatibility*: given the offered contract, the desired effort must actually be optimal for her. Competition among identical potential managers motivates a reservation-utility constraint, rather than unrestricted rent extraction.

A high profit is an imperfect signal of effort

A high realized profit can reflect high effort or good luck. Strong performance pay rewards effort but also exposes the manager to noise. The owner must trade off incentives against insurance.

2.3 Observable effort: why full insurance is optimal

Suppose effort is observable and contractible, so the owner can require e directly. Fix that effort temporarily. If profits take values Π_k with probabilities $\rho_k(e) > 0$, the owner chooses wages w_k to maximize

$$\sum_k \rho_k(e)(\Pi_k - w_k) \quad \text{subject to} \quad \sum_k \rho_k(e)u(w_k, e) \geq u_0.$$

The effort is fixed during this wage optimization; neither profits nor their distribution need be independent of effort in the overall model. Under the usual interior contracting conditions, participation binds. The Lagrangian is

$$\mathcal{L} = \sum_k \rho_k(e)(\Pi_k - w_k) + \mu \left[\sum_k \rho_k(e)u(w_k, e) - u_0 \right].$$

Differentiating with respect to w_k gives

$$-\rho_k(e) + \mu\rho_k(e)u_w(w_k, e) = 0 \quad \Longrightarrow \quad u_w(w_k, e) = \frac{1}{\mu}.$$

With strict income risk aversion, marginal utility is strictly decreasing in pay, so equal marginal utilities imply equal wages across profit states. The risk-neutral owner bears the risk; the manager is fully insured. The same argument applies with a continuous distribution.

The owner then chooses effort taking account of the insured wage needed to satisfy participation. Full insurance is feasible here because effort is contractible; a fixed wage alone would not induce costly hidden effort.

2.4 Hidden effort and a risk-neutral manager

Let $u(w, e) = w - z(e)$, with effort cost $z(e)$. Under full information, binding participation gives $\mathbb{E}w = z(e) + u_0$. The owner's effort choice is therefore

$$e^{FB} \in \arg\max_e \{\mathbb{E}\Pi(e, \varepsilon) - z(e) - u_0\}. \tag{2.4}$$

The binding constraint here is *participation*. There is no hidden-action constraint in the full-information benchmark.

Now return to hidden effort and consider a residual-claim contract

$$w(\Pi) = \Pi - K, \qquad K = \mathbb{E}\Pi(e^{FB}, \varepsilon) - z(e^{FB}) - u_0. \tag{2.5}$$

The manager's expected utility from choosing any e becomes

$$\mathbb{E}w - z(e) = \mathbb{E}\Pi(e, \varepsilon) - z(e) - K.$$

Because K does not depend on her chosen effort, the manager maximizes the same effort-dependent objective as in the first best. At e^{FB} her expected utility is u_0, and the owner receives K in every state.

Residual claimancy aligns incentives under unrestricted transfers

A risk-neutral manager can be made the residual claimant and pay a fixed fee to the owner. This implements the first-best effort and is equivalent to leasing or buying the firm for K. The manager bears all profit risk.

Why the contracting assumptions matter

The contract can require negative wages or a substantial up-front payment. Wealth constraints, limited liability or restrictions on transfers can prevent its implementation. Risk neutrality makes full risk bearing harmless; it does not make this the only possible optimal risk allocation in every contracting environment.

2.5 The opposite polar case: extreme risk aversion

Consider a manager who ranks wage lotteries primarily by their minimum guaranteed payment. Suppose the *support* of the profit distribution is the same at every effort, although effort changes probabilities. Even very high effort cannot eliminate a bad profit realization. For any compensation schedule based only on realized profit, the minimum attainable wage then does not improve with effort.

If effort is costly, the manager chooses $\underline{e}$. The owner cannot induce higher effort through this worst-outcome criterion and instead offers a constant wage satisfying participation. This conclusion uses both extreme risk aversion and the common-support assumption.

With ordinary finite risk aversion and informative but noisy performance, optimal contracts generally balance insurance and incentives; effort is often below the full-information level. A fixed component plus performance pay is a useful representation, although the optimal contract need not be linear without additional assumptions. Reputation, future earnings, product and managerial labor markets, monitoring and peer pressure can also discipline managers.

2.6 The boundaries of the firm

The make-or-buy decision compares *economic costs of internal provision* with *economic costs of external provision*, including transaction costs. A low supplier price is insufficient if incomplete contracts create delays, poor quality, bargaining problems or hold-up. Internal production can be preferable even when its narrow production cost is higher.

Horizontal scope and vertical integration can save production or distribution costs. Technologically unrelated goods may share distribution. Yet coordination costs and limited managerial span rise with organizational size. The firm's boundary reflects a balance between these savings and organizational costs. In later market models, single-product firms are often imposed for tractability; that assumption sets these organizational questions aside.

CHAPTER 3

Consumers and Demand

Lecture 1 · source PDF pp. 58–70

3.1 Behavioral assumptions

The standard consumer chooses the best available option given preferences, prices and income. This is a benchmark, rather than a claim that every observed purchase is perfectly calculated. Idiosyncratic taste shocks or nonsystematic choice errors can generate heterogeneity and smooth aggregate demand; the logit model studied later in the course is one familiar specification. Systematic behavioral biases can require a different decision model and a separate account of welfare. The lecture also contrasts anonymous market decisions with bilateral nonmarket interactions, arguing that some experimental departures from the standard benchmark are less pronounced in the former setting. Behavioral models remain a useful extension rather than being ruled out by the benchmark.

In a dynamic setting, consumers form expectations using their available information; an equilibrium may require beliefs to be consistent with outcomes. Under extrinsic uncertainty, expected utility is a common assumption. A common prior is an additional informational assumption, not part of the definition of rationality. These distinctions matter when moving from a static demand curve to learning or forward-looking demand.

3.2 From utility to a demand function

Let $q = (q_1, \ldots, q_n)$ denote quantities in the market of interest. Let q_0 be a composite of all other goods, whose price is normalized to one. A consumer with income y solves

$$\max_{q_0, q \geq 0} U(q_0, q) \quad \text{subject to} \quad q_0 + p \cdot q \leq y. \tag{3.1}$$

When $U(q_0, q) = q_0 + u(q)$ and the consumer wants more of the numeraire, the budget binds. Substituting $q_0 = y - p \cdot q$ gives

$$\max_{q \geq 0,\ p \cdot q \leq y} \{u(q) - p \cdot q + y\}. \tag{3.2}$$

If the solution has $q_0 > 0$, the affordability constraint does not bind at the margin. Interior quantities then satisfy

$$\frac{\partial u(q)}{\partial q_i} = p_i. \tag{3.3}$$

The consumer buys until marginal willingness to pay equals price. Because y is an additive constant, demand for the market goods has no income effect *within this interior region.* At a corner with $q_0 = 0$, this conclusion need not hold.

3.2.1 Unit demand and discrete choice

A consumer may buy one unit of one product, or nothing. Cars at a given purchasing occasion are the lecture's example. If the indirect utility from product i is v_i, the decision is

$$\max\{v_0, v_1, \ldots, v_n\},$$

where v_0, often normalized to zero, is the outside option. A single consumer's quantity choice is discrete. Aggregate demand can nevertheless be smooth when consumers differ in valuations, locations or tastes.

3.3 A linear-quadratic model with two goods

Consider

$$U(q_0, q_1, q_2) = q_0 + a(q_1 + q_2) - \frac{1}{2}\left(bq_1^2 + 2dq_1q_2 + bq_2^2\right), \qquad b > |d|. \tag{3.4}$$

The Hessian of utility in (q_1, q_2) is

$$-\begin{pmatrix} b & d \\ d & b \end{pmatrix}.$$

Its eigenvalues are $-(b+d)$ and $-(b-d)$, both negative under $b > |d|$. Utility is strictly concave, so any feasible interior first-order solution is the unique optimum.

After substituting the budget constraint, differentiate with respect to the two quantities:

$$p_1 = a - bq_1 - dq_2, \tag{3.5}$$
$$p_2 = a - dq_1 - bq_2. \tag{3.6}$$

These are the inverse demands. To recover direct demands, write

$$\begin{pmatrix} b & d \\ d & b \end{pmatrix}\begin{pmatrix} q_1 \\ q_2 \end{pmatrix} = \begin{pmatrix} a - p_1 \\ a - p_2 \end{pmatrix}, \qquad \begin{pmatrix} b & d \\ d & b \end{pmatrix}^{-1} = \frac{1}{b^2 - d^2}\begin{pmatrix} b & -d \\ -d & b \end{pmatrix}.$$

Multiplying out,

$$q_1 = \frac{a(b-d) - bp_1 + dp_2}{b^2 - d^2} = \tilde{a} - \tilde{b}p_1 + \tilde{d}p_2, \tag{3.7}$$
$$q_2 = \frac{a(b-d) + dp_1 - bp_2}{b^2 - d^2} = \tilde{a} + \tilde{d}p_1 - \tilde{b}p_2, \tag{3.8}$$

with

$$\tilde{a} = \frac{a}{b+d}, \qquad \tilde{b} = \frac{b}{b^2 - d^2} > 0, \qquad \tilde{d} = \frac{d}{b^2 - d^2}. \tag{3.9}$$

What the cross term means

For interior demand, $\partial q_i / \partial p_j = \tilde{d}$. Goods are substitutes if $d > 0$, independent if $d = 0$, and complements if $d < 0$. Inverse and direct demand describe the same preferences; their coefficients are different because one system is the inverse of the other.

Nonnegative consumption requires a corner analysis

These linear equations describe the region where both goods are consumed and the numeraire is positive. If a calculated quantity is negative, set that good to zero and re-solve the remaining optimization problem. Simply truncating each equation separately can give the wrong demand for the other good because the goods interact.

For example, on a boundary with $q_2 = 0$, the first good satisfies $q_1 = [(a - p_1)/b]_+$; this boundary is valid only if the excluded good's marginal net utility is nonpositive: $a - dq_1 - p_2 \leq 0$.

3.4 Individual and aggregate demand

With N identical consumers, aggregate demand is N times individual demand. A representative-consumer model is useful when this scaling is acceptable. Alternatively, let consumer type θ have distribution F and individual demand $q_i(p;\theta)$. Aggregate demand is

$$Q_i(p) = M \int q_i(p;\theta)\,\mathrm{d}F(\theta).$$

Even if every individual chooses either zero or one unit, a continuous distribution of switching thresholds can generate a continuous downward-sloping Q_i. The Hotelling model later in the lecture provides precisely this link between heterogeneous tastes and aggregate demand.

CHAPTER 4

Welfare in a Partial-Equilibrium Model

Lecture 1 · source PDF pp. 71–83

4.1 What is held fixed?

A partial-equilibrium analysis concentrates on one market, holding input prices, income and conditions in other markets fixed. It is a useful approximation when the market is small relative to the economy and spillovers to other markets are limited. General equilibrium would instead determine these other prices and incomes jointly.

Quasilinear utility gives an especially clean local foundation: when the numeraire remains positive, market demand is independent of income. Cobb–Douglas preferences do *not* generally remove income effects; a good with expenditure share α has demand $q = \alpha y/p$. A small expenditure share may justify an approximation, but the functional form alone does not make ordinary consumer surplus an exact welfare measure.

4.2 Consumer surplus for one good

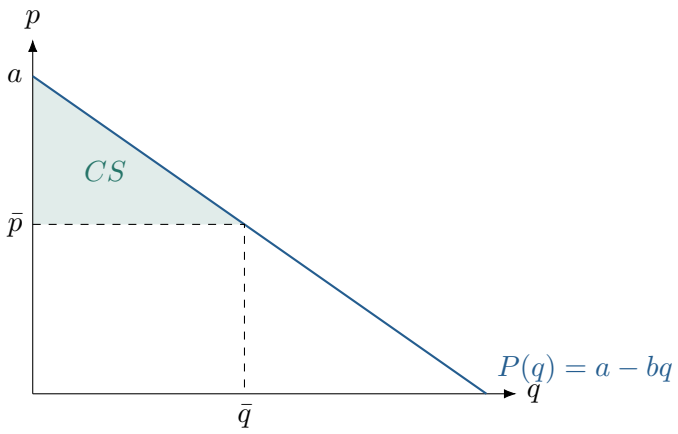
For inverse demand $P(q)$ and consumption $\bar{q}$ at price $\bar{p} = P(\bar{q})$, consumer surplus is

$$\mathrm{CS} = \int_0^{\bar{q}} P(z)\,\mathrm{d}z - \bar{p}\bar{q}. \tag{4.1}$$

The integral is gross willingness to pay; the second term is expenditure. With $P(q) = a - bq$,

$$\mathrm{CS} = a\bar{q} - \frac{b}{2}\bar{q}^2 - (a - b\bar{q})\bar{q} = \frac{b}{2}\bar{q}^2. \tag{4.2}$$

Geometrically it is the triangle below demand and above the horizontal price line.



An area can be drawn even when it is not exact welfare

Ordinary demand reflects income effects as well as substitution. The area under it is an exact money measure of welfare change under suitable quasilinear conditions. Otherwise it is generally an approximation; compensating or equivalent variation uses expenditure functions and compensated demand. A small market makes the neglected income effects less consequential.

4.3 Consumer surplus with several goods

For quasilinear utility $q_0 + u(q)$ and an interior choice,

$$\mathrm{CS}(q) = u(q) - p \cdot q. \tag{4.3}$$

For the quadratic model of Section 3.3, substitute inverse demand into expenditure:

$$p_1 q_1 + p_2 q_2 = a(q_1 + q_2) - b(q_1^2 + q_2^2) - 2dq_1 q_2.$$

Subtracting from gross utility yields

$$\boxed{\mathrm{CS}(q_1, q_2) = \frac{1}{2}\left(bq_1^2 + 2dq_1 q_2 + bq_2^2\right).} \tag{4.4}$$

Cross-price derivatives satisfy $\partial q_1/\partial p_2 = \partial q_2/\partial p_1 = \tilde{d}$, consistent with an underlying quasilinear surplus function. This symmetry helps make a simultaneous multi-price welfare change well defined rather than dependent on the order of integration.

As $d = b$, utility depends on $Q = q_1 + q_2$ through $aQ - bQ^2/2$, and Equation (4.4) becomes $\mathrm{CS} = bQ^2/2$. This is the homogeneous-good limit: the consumer is indifferent between equal-priced units of the two products. The direct-demand matrix becomes singular at the limit, so one should use aggregate demand rather than invert that matrix.

4.4 Producer surplus, total surplus and distribution

For a collection of firms, operating producer surplus is revenue less variable production cost. Profit additionally deducts relevant fixed costs. The two have the same *change* when fixed costs are unchanged, but are not universally identical in levels.

With all relevant resource costs included consistently, total surplus is

$$\mathrm{TS} = \mathrm{CS} + \sum_i \pi_i. \tag{4.5}$$

Prices paid by consumers are received by firms, so those transfers cancel. In a single-good setting, this leaves

$$\mathrm{TS}(Q) = \int_0^Q P(z)\,\mathrm{d}z - C(Q).$$

The efficient interior quantity satisfies $P(Q) = MC(Q)$: marginal willingness to pay equals marginal resource cost.

Adding surpluses assigns equal weight to a dollar received by a consumer and a dollar received by a producer. It also abstracts from distributional differences among consumers. Without

quasilinearity or fixed welfare weights, redistributing income can affect both demand and the interpretation of aggregate surplus.

The lecture contrasts a consumer-welfare standard, concerned with consumer benefits, and a total-welfare standard, concerned with their sum with producer benefits. It cites the European Commission's 1997 competition-policy discussion as a historical illustration of emphasis on consumers. The precise legal welfare standard depends on the jurisdiction and context; the analytical distinction is that a pure transfer to producers can leave total surplus unchanged while reducing consumer surplus.

CHAPTER 5

Competition and Monopoly

Lecture 1 · source PDF pp. 84–113

5.1 The perfectly competitive benchmark

A price-taking firm treats the market price p as fixed when it changes its output. This can approximate an industry with many firms, each small relative to market demand. Industry supply and demand determine the price collectively, even though no individual firm treats its own effect on price as material.

The firm's perceived demand is horizontal. It solves

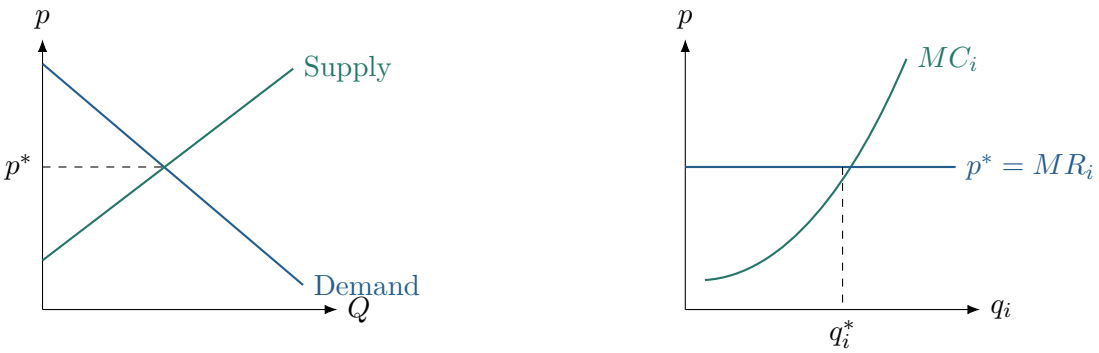
$$\max_{q \geq 0} \{pq - C(q)\}.$$

At an interior optimum, $p = C'(q)$. If marginal cost rises, an increase in price raises the chosen quantity. The rising portion of marginal cost is the firm's supply curve *where production is preferable to shutdown*.

Competitive output

Conditional on an interior operating optimum, a competitive firm produces where price equals marginal cost. With convex variable costs, its supply is increasing in price. For a U-shaped variable-cost curve, the familiar short-run supply curve is the rising marginal-cost segment above minimum average variable cost, when fixed cost is unavoidable.

A competitive firm can earn positive operating surplus when marginal cost rises: earlier units cost less than the last unit. Zero economic profit is an additional long-run free-entry conclusion under suitable assumptions, not the same statement as $p = MC$.



The left panel determines a market price. The right panel shows how an individual firm takes that price as given. The two demand curves answer different questions.

5.2 A monopolist's choice

A monopolist is the only supplier in the market being modeled. Let $P'(q) < 0$ and costs be $C(q)$. It chooses

$$\max_{q \geq 0} \pi(q) = qP(q) - C(q). \tag{5.1}$$

Differentiating the *product* $qP(q)$ gives

$$\pi'(q) = P(q) + qP'(q) - C'(q) = 0. \tag{5.2}$$

The term $P(q)$ is revenue from an extra unit at the current price. The term $qP'(q) < 0$ is the loss on existing units caused by lowering price to sell that extra unit. Thus marginal revenue is below price.

An interior first-order solution is a strict maximum if

$$\pi''(q) = 2P'(q) + qP''(q) - C''(q) < 0.$$

Concave revenue and convex cost are sufficient for concave profits; strict concavity gives uniqueness if an optimum exists. A downward-sloping concave inverse demand and convex costs are convenient sufficient conditions, but not necessary ones. Boundary choices and the option not to operate must still be checked.

5.2.1 Deriving the inverse-elasticity rule

Rearrange Equation (5.2):

$$P(q) - C'(q) = -qP'(q).$$

If demand and inverse demand are locally invertible, $Q'(p) = 1/P'(q)$ and

$$\eta = -Q'(p)\frac{p}{q} = -\frac{P(q)}{qP'(q)}.$$

Dividing the first-order condition by price yields

$$\boxed{L \equiv \frac{p - MC}{p} = \frac{1}{\eta}.} \tag{5.3}$$

The left side is the Lerner index: the price-cost difference as a fraction of *price*, rather than cost. Less elastic demand supports a higher markup, evaluated at the optimum.

A positive-cost interior monopolist operates on elastic demand

If $p > 0$ and $MC > 0$, the inverse-elasticity rule requires $\eta > 1$. With $MC = 0$, an interior revenue maximum has $\eta = 1$. A monopolist with nonnegative marginal cost does not optimally choose a regular interior point with $\eta < 1$: marginal revenue would be negative, so reducing output would raise revenue and weakly reduce cost. Extrapolating $1/\eta$ toward infinity is not a valid interior-optimum prediction.

As demand becomes very elastic, the Lerner index tends toward zero. This is a statement about the price-cost margin; it does not by itself prove that total profit is zero when fixed costs or inframarginal cost savings are present.

The same monopoly allocation can be found by choosing price:

$$\max_{p}\{pQ(p) - C(Q(p))\}, \qquad Q(p) + (p - MC)Q'(p) = 0.$$

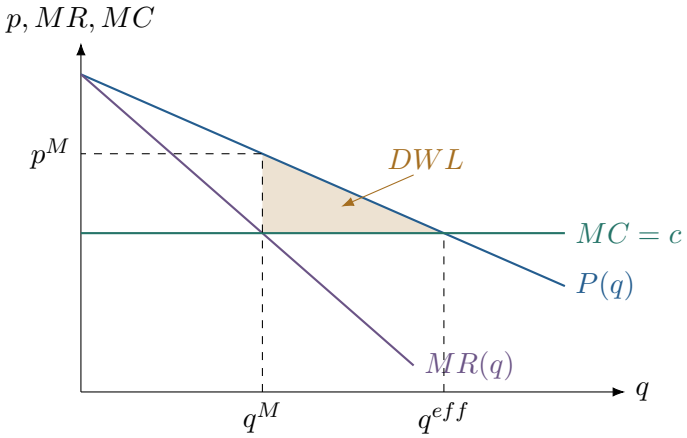
Where P and Q describe the same one-to-one feasible demand relationship, changing the choice variable does not change the optimum. This equivalence will not carry over to changing rivals' strategic commitments in an oligopoly.

Linear demand: a complete monopoly calculation

Let $P(q) = a - bq$, $a > c \geq 0$, and $C(q) = F + cq$ when operating. Then

$$a - 2bq - c = 0 \quad \Longrightarrow \quad q^M = \frac{a-c}{2b}, \quad p^M = \frac{a+c}{2}.$$

Operating profit is $(a-c)^2/(4b)$; total profit is this amount minus F. If F is avoidable, production is chosen only when the operating profit covers it. Consumer surplus is $(a-c)^2/(8b)$. The efficient output is $q^{eff} = (a-c)/b$, and the deadweight loss from monopoly is $(a-c)^2/(8b)$, when the same fixed costs are relevant in both allocations.



5.3 Multiproduct monopoly: the general condition

A two-product firm chooses (p_1, p_2) and faces quantities $Q_1(p_1, p_2)$ and $Q_2(p_1, p_2)$. Its cost is $C(Q_1, Q_2)$. Let $C_i = \partial C/\partial q_i$ and $Q_{j,i} = \partial Q_j/\partial p_i$. Profit is

$$\pi = p_1 Q_1 + p_2 Q_2 - C(Q_1, Q_2).$$

The chain rule gives, for $i \neq j$,

$$\begin{aligned} 0 &= Q_i + p_i Q_{i,i} + p_j Q_{j,i} - C_i Q_{i,i} - C_j Q_{j,i} && (5.4) \\ &= Q_i + (p_i - C_i) Q_{i,i} + (p_j - C_j) Q_{j,i}. && (5.5) \end{aligned}$$

The last term is the extra margin gained or lost on the *other* product when p_i changes. A single-product firm does not internalize this term.

5.3.1 Linked demands, separable costs

Suppose $C(q_1, q_2) = C_1(q_1) + C_2(q_2)$ and $Q_{i,i} < 0$. With own-price elasticity $\eta_i = -p_i Q_{i,i}/Q_i$, solve Equation (5.5) for product i's markup:

$$\frac{p_i - C_i'}{p_i} = \frac{1}{\eta_i} + \frac{p_j - C_j'}{p_i} \frac{Q_{j,i}}{-Q_{i,i}}. \tag{5.6}$$

If products are substitutes, $Q_{j,i} > 0$: a price rise on i diverts customers to j. Joint ownership internalizes this recovered profit and, under positive margins and the usual regularity conditions, weakens the incentive to cut price. If they are complements, $Q_{j,i} < 0$: raising p_i harms the other product's sales, so joint ownership adds an incentive to lower p_i.

Internalizing substitution or complementarity

For substitutes, a lost sale may be recaptured by another product of the same firm. For complements, a lower price can stimulate profitable sales elsewhere in the product line. A complementary product can even be priced below marginal cost when its indirect benefits are large enough; free software or demonstrations illustrate the broader cross-product logic.

Equation (5.6) is a local incentive statement. A global ranking of all equilibrium prices requires further regularity and a well-specified comparison with separate ownership; interacting demand and cost changes can otherwise overturn simple rankings.

5.3.2 Linked costs, independent demands

Suppose instead $Q_i = Q_i(p_i)$, so $Q_{j,i} = 0$. The first-order condition becomes

$$\frac{p_i - C_i(q_i, q_j)}{p_i} = \frac{1}{\eta_i}. \tag{5.7}$$

The rule has its familiar form, but marginal cost depends on the other output. If $C_{ij} < 0$, an increase in q_j reduces the marginal cost of product i, encouraging additional production and a lower price. A joint firm can exploit this cost complementarity, while separate production may face $C_i(q_i, 0)$. Diseconomies reverse this force.

If neither demand nor cost is linked, the maximization separates into two monopoly problems. The highest relative markup then occurs on the product with the least elastic demand. With both kinds of linkage present, retain every term in Equation (5.5) rather than applying a one-product formula mechanically.

CHAPTER 6

A Dominant Firm and a Competitive Fringe

Lecture 1 · source PDF pp. 114–136

6.1 The economic idea and the lecture case

A dominant firm has market power but faces a set of small price-taking suppliers. Their supply behavior limits its residual demand. Unlike a pure monopoly, the dominant firm's demand may depend on competitors' *costs* as well as on consumer tastes.

Lecture case: branded drugs and generics

The slides discuss Merck's Zocor after patent expiry and the 180-day generic exclusivity granted to Teva and Ranbaxy. They describe a sharp Merck price cut in this episode. The broader mechanism is that generic firms can compete closely with one another while an established brand remains differentiated and can retain a premium. A branded firm's price response to expiry is not universally a decrease; segmentation and consumer substitution matter.

6.2 Hotelling demand with two endpoints

Products 1 and 2 are located at 0 and 1. A mass M of consumers is uniformly distributed over $x \in [0,1]$. Each buys at most one unit and obtains

$$v_i(x) = r - \tau|x - l_i| - p_i, \qquad \tau > 0,$$

with outside-option utility zero. The parameter τ is the cost of distance or taste mismatch. A larger τ makes a distant alternative less attractive.

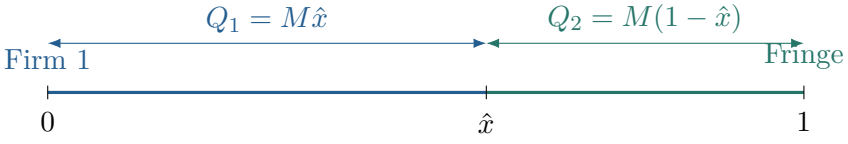
In a covered market with both products consumed, the indifferent consumer solves

$$r - \tau\hat{x} - p_1 = r - \tau(1 - \hat{x}) - p_2 \quad \Longrightarrow \quad \hat{x} = \frac{1}{2} + \frac{p_2 - p_1}{2\tau}.$$

All consumers to the left buy product 1; those to the right buy product 2. Hence

$$Q_i(p_i, p_j) = M\left(\frac{1}{2} + \frac{p_j - p_i}{2\tau}\right). \tag{6.1}$$

This expression requires $0 < \hat{x} < 1$ and nonnegative utility for the purchasing consumers. At uncovered or corner allocations, the demand expression changes.



6.3 Constant marginal costs

Let both segments have constant marginal cost c. Competition in product 2 gives $p_2 = c$, independently of p_1. Firm 1 solves

$$\max_{p_1}(p_1 - c)M\left(\frac{1}{2} + \frac{c - p_1}{2\tau}\right).$$

Differentiation gives

$$\frac{M}{2\tau}(\tau + 2c - 2p_1) = 0,$$

so

$$p_1^* = c + \frac{\tau}{2}, \qquad Q_1^* = \frac{M}{4}, \qquad \pi_1^* = \frac{M\tau}{8}. \tag{6.2}$$

The dominant firm charges a markup while the fringe charges marginal cost. These results assume valuations are high enough to cover the market at these prices. A higher valuation for the branded product would change the residual-demand intercept.

6.4 Increasing marginal costs: the fringe reacts

Now let segment costs be $C_i(q_i) = cq_i^2/2$, with $c > 0$. The competitive fringe clears where its price equals marginal cost:

$$p_2 = cQ_2(p_1, p_2) = cM\left(\frac{1}{2} + \frac{p_1 - p_2}{2\tau}\right).$$

Collecting the terms involving p_2,

$$p_2(p_1) = \frac{cM(\tau + p_1)}{2\tau + cM}, \qquad p_2'(p_1) = \frac{cM}{2\tau + cM} > 0. \tag{6.3}$$

This is a *market-clearing response*, sometimes called a pseudo best response. It is not the choice of a single strategic fringe player. The dominant firm anticipates the whole schedule.

Substitute Equation (6.3) into its demand:

$$Q_1(p_1, p_2(p_1)) = \frac{M}{2\tau}\left[\tau - p_1 + \frac{cM(\tau + p_1)}{2\tau + cM}\right] \tag{6.4}$$

$$= \frac{M}{2\tau + cM}(\tau + cM - p_1). \tag{6.5}$$

The cost parameter c now enters the dominant firm's *demand*. Changing industry technology affects both its own costs and the supply of alternatives.

To solve the pricing problem cleanly, let

$$A = \tau + cM, \qquad B = \frac{2\tau + cM}{M}.$$

Equation (6.5) implies $p_1 = A - Bq_1$. The dominant firm maximizes

$$\pi_1(q_1) = Aq_1 - \left(B + \frac{c}{2}\right)q_1^2.$$

Its first-order condition is $A - (2B + c)q_1 = 0$. Substituting back,

Dominant-firm outcome with increasing marginal costs

$$Q_1^* = \frac{M(\tau + cM)}{4\tau + 3cM},$$
$$p_1^* = \frac{2(\tau + cM)^2}{4\tau + 3cM},$$
$$\pi_1^* = \frac{M(\tau + cM)^2}{2(4\tau + 3cM)}.$$

The fringe price follows by substituting p_1^* into Equation (6.3).

6.4.1 Why higher industry costs can raise profit

Differentiate the equilibrium profit with respect to c:

$$\frac{\partial \pi_1^*}{\partial c} = \frac{M^2(\tau + cM)(5\tau + 3cM)}{2(4\tau + 3cM)^2} > 0. \tag{6.6}$$

Also,

$$\frac{\partial p_1^*}{\partial c} = \frac{2M(\tau + cM)(5\tau + 3cM)}{(4\tau + 3cM)^2} > 0.$$

Holding residual demand fixed, a rise in own cost hurts profit. But an industry-wide rise in c rotates fringe supply upward and makes the alternative more expensive. Here this demand-shift benefit dominates the own-cost loss. Fringe producer surplus is $p_2 Q_2 - cQ_2^2/2 = cQ_2^2/2$, and also rises in this particular covered-market model.

An own-cost effect and a rival-supply effect

$$c \uparrow \Longrightarrow \begin{cases} \text{own production becomes more costly,} \\ \text{fringe supply becomes less competitive.} \end{cases}$$

The second channel can outweigh the first. The direction of profit change must be established in the specified model; it is not a general benefit from higher costs.

Even with competitive pricing at both locations, a steeper common marginal-cost schedule can increase producer surplus on inframarginal units. This anticipates the later distinction between competition intensity and measured profit levels.

CHAPTER 7

Market Definition and Market Performance

Lecture 1 · source PDF pp. 137–167

7.1 From isolated decisions to strategic interaction

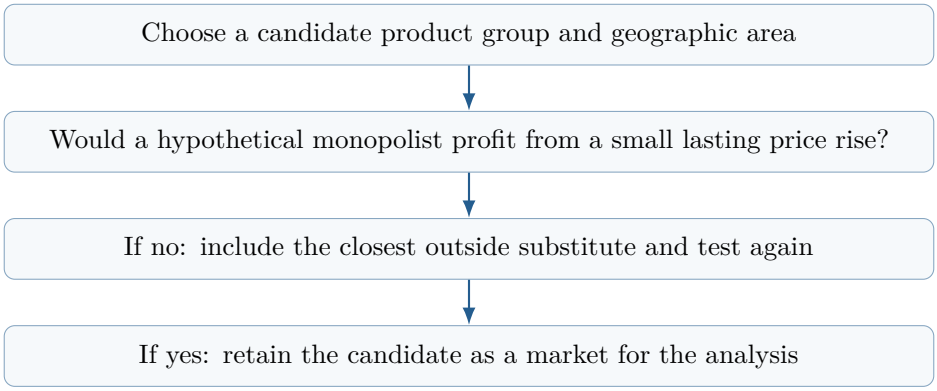
A competitive firm treats price as given; a monopolist treats the demand curve as given; a dominant firm can optimize against an anticipated fringe-supply schedule. When several strategic firms affect one another, each must also account for the fact that rivals optimize. Nash equilibrium requires mutual optimality. This is the main subject of Lecture 2.

Before measuring market power, define the market in which a firm operates. Otherwise both its competitors and the denominator of its market share are ambiguous.

7.2 Demand substitution and the hypothetical monopolist

A relevant market has a product dimension and a geographical dimension; timing can also matter. Start with the product of interest and its closest substitutes, which impose the strongest competitive constraints.

The hypothetical-monopolist test asks whether a hypothetical owner of a candidate group of products could profitably impose a small, significant and nontransitory price increase, holding terms of sale outside the candidate group fixed. A 5–10% increase is a common illustrative magnitude. If too many consumers switch outside, expand the candidate group and repeat. Stop when the candidate group can sustain the increase.



The baseline and the period are part of the test

For a merger assessment, prevailing prices are commonly a starting point unless the facts call for another benchmark. In an existing-market-power investigation, an already elevated price can produce misleading substitution evidence, the classic cellophane problem. A one-year duration is a classroom illustration rather than a universal rule. The 2024 European Commission Market Definition Notice explains the context-dependent methodology; see the references.

7.3 Supply substitution and entry

Consumers need not regard two goods as substitutes for their producers to constrain one another. Suppliers may be able to switch existing production quickly, effectively and without substantial sunk costs. Such immediate supply substitution can justify a broader market. More remote potential entry is generally evaluated in the competitive assessment rather than automatically included in the market definition.

Lecture case: paper grades and Torras/Sarrio

Readers may view different coated paper grades as poor substitutes. Paper producers, however, can sometimes switch between grades using the same plant and raw materials. A hypothetical monopolist of one grade may then face prompt supply from firms producing other grades. The lecture's merger case illustrates why market boundaries can be broader than demand substitution alone suggests.

This distinction matters: an existing producer's prompt switch differs from a new entrant building a factory. Both can constrain profits, but they operate on different time horizons.

7.4 The Lerner index

Define

$$L_i = \frac{p_i - MC_i}{p_i}. \tag{7.1}$$

It measures a static price-cost margin, using *economic marginal cost*. Accounting average cost is not a substitute. Marginal costs are often unobserved, so measuring L_i may require a structural demand-and-conduct model.

A low current price does not by itself establish strong competition. A firm may sacrifice current margins for future gains, or price a complementary product cheaply to earn elsewhere. Conversely, high operating profits can reflect low costs, scarce inputs or fixed investments. Lecture 2 develops models that link observed outcomes to conduct under explicit assumptions.

7.5 Concentration ratios and the Herfindahl index

Order market shares $s_1 \geq s_2 \geq \cdots \geq s_n$, with $\sum_i s_i = 1$. The m-firm concentration ratio and Herfindahl index are

$$CR_m = \sum_{i=1}^{m} s_i, \qquad H = \sum_{i=1}^{n} s_i^2. \tag{7.2}$$

The concentration ratio summarizes the leading firms but discards the distribution within and outside that group. The HHI uses every squared share and gives larger firms more weight.

For a fixed n, write $s_i = 1/n + (s_i - 1/n)$ and expand:

$$H = \frac{1}{n} + \sum_i \left(s_i - \frac{1}{n}\right)^2. \tag{7.3}$$

Thus $H \geq 1/n$, with equality for equal shares. Greater inequality raises concentration. A monopoly has $H = 1$. If two firms with shares s_i, s_j merge and shares otherwise remain fixed,

$$\Delta H = (s_i + s_j)^2 - s_i^2 - s_j^2 = 2s_i s_j > 0.$$

Using percentage shares instead multiplies the normalized index by 10,000; the corresponding monopoly value is 10,000.

Same firm count, different concentration

Four equal firms have $H = 4(0.25)^2 = 0.25$, or 2,500 points. Shares $(0.7, 0.1, 0.1, 0.1)$ give $H = 0.52$, or 5,200 points. The firm count is identical; the distribution is much less balanced in the second market.

7.5.1 Screening rules require context

Concentration is a screening statistic, not a universal measure of market power. In the standard symmetric homogeneous Bertrand model, two firms give an HHI of 5,000 points but a zero equilibrium price-cost margin. Capacity constraints and product differentiation can produce very different implications at the same concentration.

The 2010 US merger guidelines used the historically cited 1,500 and 2,500 point boundaries. In the 2023 DOJ/FTC Merger Guidelines, a post-merger HHI above 1,800 together with an increase above 100 points raises a structural presumption under Guideline 1; the guidelines also describe a separate share-based presumption. These are rebuttable assessment frameworks, not an economic identity or an automatic conclusion for every merger. In EU case law, a sustained 50% share can be strong presumptive evidence of dominance, subject to context and exceptional circumstances; there is no universal definition that declares every firm dominant solely upon crossing one share threshold.

7.6 Lecture case: concentration across manufacturing industries

The table reproduces the historical figures in the supplied slides. They concern the largest 50 companies within broad US manufacturing industry groups in 1997 and 2002. These broad groups need not coincide with economically relevant antitrust markets, and a top-50 calculation can omit smaller firms' squared shares.

Industry group	1997	2002	Change
Transportation equipment	797.6	574.7	−222.9
Beverage and tobacco products	777.2	709.5	−67.7
Petroleum and coal products	350.0	543.4	+193.4
Textile product mills	186.2	403.0	+216.8
Paper	173.3	259.3	+86.0
Leather and allied products	167.2	176.6	+9.4
Computer and electronic products	136.6	135.0	−1.6
Electrical equipment, appliances and components	105.9	113.9	+8.0
Apparel	100.6	105.7	+5.1
Primary metals	97.4	149.6	+52.2
Textile mills	94.4	105.6	+11.2
Food	91.0	118.7	+27.7
Chemicals	76.6	99.9	+23.3
Furniture and related products	55.5	57.2	+1.7
Machinery	55.4	71.3	+15.9
Wood products	52.7	48.4	−4.3
Nonmetallic mineral products	52.1	46.7	−5.4
Printing and related support activities	38.4	45.2	+6.8
Plastics and rubber products	30.2	32.0	+1.8
Fabricated metal products	8.5	10.2	+1.7

The substantive lesson is variation: concentration differs greatly across industries and can rise in one while falling in another. A link from HHI to markups requires a model. Under homogeneous Cournot competition, Lecture 2 derives the share-weighted relation $\bar{L} = H/\eta$; elasticity and conduct are essential parts of that result.

APPENDIX A

A Compact Formula Reference

The formulas below use the assumptions established in the text. They are a retrieval aid, rather than a replacement for those assumptions.

Object	Formula or condition
Average-cost slope	$\mathrm{AC}'(q) = (\mathrm{MC}(q) - \mathrm{AC}(q))/q$.
Fixed plus linear cost	$C(q) = F + cq$, $\mathrm{AC} = c + F/q$, $\mathrm{MC} = c$.
Manager participation	$\mathbb{E}u(w(\Pi(e^*, \varepsilon)), e^*) \geq u_0$.
Manager incentives	$e^* \in \arg\max_e \mathbb{E}u(w(\Pi(e, \varepsilon)), e)$.
Residual claimant	$w(\Pi) = \Pi - K$, implementing first-best effort for a risk-neutral manager with unrestricted transfers.
Quadratic inverse demand	$p_i = a - bq_i - dq_j$, $b > \lvert d \rvert$.
Quadratic direct demand	$q_i = \dfrac{a(b-d) - bp_i + dp_j}{b^2 - d^2}$, interior.
Quadratic consumer surplus	$\mathrm{CS} = \frac{1}{2}[b(q_1^2 + q_2^2) + 2dq_1q_2]$.
Competition	$p = MC$ at an interior operating optimum.
Monopoly	$P(q) + qP'(q) = C'(q)$; $L = 1/\eta$.
Linear monopoly	$q^M = (a-c)/(2b)$, $p^M = (a+c)/2$.
Multiproduct condition	$Q_i + (p_i - C_i)Q_{i,i} + (p_j - C_j)Q_{j,i} = 0$.
Covered Hotelling demand	$Q_i = M[1/2 + (p_j - p_i)/(2\tau)]$.
Constant-cost dominant firm	$p_1 = c + \tau/2$, $Q_1 = M/4$, $\pi_1 = M\tau/8$.
Competitive fringe schedule	$p_2(p_1) = cM(\tau + p_1)/(2\tau + cM)$ for $C_i = cq_i^2/2$.
Concentration	$CR_m = \sum_{i=1}^{m} s_i$, $H = \sum_i s_i^2$.
HHI dispersion identity	$H = 1/n + \sum_i (s_i - 1/n)^2$.
Merger HHI change	$\Delta H = 2s_is_j$ with other shares fixed.

APPENDIX B

Source Concordance

The supplied file *1-firms, consumers and the market_v1.pdf* has 167 physical pages. The printed slide counters repeat across progressive builds. This table includes all physical pages; title pages and repeated outlines are represented by the reading guide and the corresponding chapter structure.

PDF pages	Source material	Destination
1–6	Title, introduction, objectives, overview and division	Reading guide; chapter structure
7–10	Firms, consumers and the black-box representation	Chapters 1–3
11–13	Opportunity costs and cost function	Chapter 1, economic costs
14–17	Economies of scale and scope	Chapter 1, scale and scope
18–21	Fixed and sunk costs	Chapter 1, decision timing
22–28	Profit maximization, alternatives and outline	Chapter 1, objectives; Chapter 2 opening
29–38	Organization, principal–agent model, IR and IC	Chapter 2, model and constraints
39–44	Full-information insurance benchmark	Chapter 2, observable effort
45–49	Risk-neutral manager and residual-claim contract	Chapter 2, hidden effort
50–53	Extreme risk aversion, lessons and other discipline	Chapter 2, insurance and incentives
54–58	Make-or-buy, horizontal/vertical scope and outline	Chapter 2, firm boundaries
59–62	Rationality, biases, expectations and uncertainty	Chapter 3, behavioral assumptions
63–66	Utility, budget, quasilinearity and unit demand	Chapter 3, individual choice
67–68	Quadratic utility and inversion of demand	Chapter 3, linear demand derivation
69–71	Representative/heterogeneous consumers and outline	Chapter 3, aggregation
72–73	Partial versus general equilibrium	Chapter 4, scope of the analysis
74–78	Welfare measures and single-good CS figure	Chapter 4, one-good surplus and diagram
79–81	Multiproduct CS and homogeneous-good limit	Chapter 4, multi-good surplus
82–86	Distribution, welfare standards and transition	Chapter 4 conclusion; Chapter 5 opening
87–92	Competition, perceived demand and supply figures	Chapter 5, competitive benchmark and diagrams
93–100	Monopoly, concavity, elasticity and choice variable	Chapter 5, monopoly derivation and diagram
101–104	General multiproduct pricing and linked demand	Chapter 5, multiproduct first-order condition
105–108	Substitutes, complements and below-cost pricing	Chapter 5, linked-demand incentives
109–114	Linked costs, scope and pricing lesson	Chapter 5, cost complementarity

PDF pages	Source material	Destination
115–118	Dominant firm and Zocor/generics case	Chapter 6 opening and lecture case
119–125	Hotelling demand and competitive fringe setup	Chapter 6, environment and demand
126–127	Constant marginal costs	Chapter 6, constant-cost result
128–132	Rising costs, fringe response and optimization	Chapter 6, residual demand and solution
133–136	Industry cost comparative statics and profit	Chapter 6, two cost channels
137–143	Strategic interaction and market definition overview	Chapter 7 opening
144–148	Market boundaries, SSNIP and supply substitution	Chapter 7, definition and substitution
149–152	Torras/Sarrio paper-grade case and outline	Chapter 7, supply-side case
153–155	Lerner index, costs and dynamic limitations	Chapter 7, Lerner index
156–162	Concentration ratios, HHI and screening rules	Chapter 7, concentration and qualifications
163–165	Manufacturing HHI case and both table panels	Chapter 7, historical table (all 20 rows)
166–167	Limits of HHI and link to the next lecture	Chapter 7 conclusion

Handwritten emphasis is incorporated into the explanations of opportunity cost, fixed versus sunk costs, demand versus inverse demand, discrete choice, the signs of d, marginal revenue, elasticity, joint pricing and the cost link across products. The text explains these ideas in complete form; decorative marks and unreadable fragments are not treated as additional factual claims.

Sources and Further Reading

Primary course source. Hui Shen. *Firms, Consumers and the Market*. Advanced Industrial Organization lecture slides, September 4, 2026. Supplied annotated file: `1-firms, consumers and the market_v1.pdf`, 167 physical pages. All classroom cases and the manufacturing concentration table are taken from this course material.

Market definition. European Commission (2024). *Commission Notice on the definition of the relevant market for the purposes of Union competition law*, C/2024/1645. This supplies the qualification distinguishing immediate supply substitution from more remote entry and the context dependence of the hypothetical-monopolist test. Official notice.

Concentration screening. US Department of Justice and Federal Trade Commission (2023). *Merger Guidelines*, Guideline 1. The 1,800/100-point structural presumption and its rebuttable character are discussed here. The historical 2010 thresholds cited in the course are identified as historical. Official Guideline 1.

Dominance and market shares. Court of Justice, *AKZO Chemie BV v Commission*, Case C-62/86, judgment of July 3, 1991, paragraph 60. Large shares are evidence assessed in context, with a presumption subject to exceptional circumstances, rather than a universal mechanical definition. Official case record.

The economic models are presented for study. Historical forecasts, old industry structures and historical case events do not constitute updated market descriptions. Policy statements are attributed to the dated sources above.